THE LOCAL NEIGHBORLINESS THRESHOLD OF THE SYMMETRIC MOMENT CURVE: AN INTEGRAL CRITERION, SHARP ASYMPTOTICS, AND THE BARVINOK–LEE–NOVIK CONJECTURE FOR $k \le 440$ AND FOR ALL SUFFICIENTLY LARGE k

ABSTRACT. Let ϕ_k be the largest ϕ such that any k points of the symmetric moment curve $U_k(t) = (\cos t, \sin t, \cos 3t, \sin 3t, \dots, \cos(2k-1)t, \sin(2k-1)t)$ lying on an open arc of length ϕ span a face of $\operatorname{conv} U_k$. Barvinok, Lee and Novik proved $\pi/2 < \phi_k < \pi$ and $\phi_k \to \pi/2$, reduced ϕ_k to configurations collapsed to the two endpoints of the arc, and conjectured that the balanced collapse is always the worst one. We turn their characterization into an explicit two-integral criterion. We use it to prove: $\phi_k < \frac{\pi}{2} + \frac{\log(8\pi(k+1/2))}{4(k-1)}$ for even $k \ge 4$; $\phi_k < \frac{\pi}{2} + \frac{\log(64\sqrt{\pi}(k-1)\sqrt{k-1/2})}{2k-8}$ for odd $k \ge 13$; and $\liminf_k \left(4k(\phi_k - \pi/2) - \log k\right) \ge 2\log 2$. We also prove the effective two-sided bound $\frac{\pi}{2} + \frac{\log k}{4k} < \phi_k < \frac{\pi}{2} + \frac{\log k + 4.1}{4k}$ for every $k \ge 2$; on a finite range of k it rests on explicit closed-form inequalities certified in ball arithmetic. Refining the analysis we obtain the exact second-order term, $\lim_k \left(4k(\phi_k - \frac{\pi}{2}) - \log k\right) = \log(8\pi)$, which is the asymptotics of the balanced split. We prove the Barvinok–Lee–Novik conjecture for all $k \le 440$ (both parities; computer-assisted, ball arithmetic), together with the monotonicity of ϕ_k in that range. We also prove it, analytically but with an ineffective threshold, for all sufficiently large k. The splits far from balanced are excluded by the sharp asymptotics. For near-balanced splits we show that the criterion is strictly concave in the exponent: its second derivative is at most $-4\delta/N + O(N^{-2})$ with $N\delta \asymp \log k$. This is proved with Stein identities for the weights $\sin^a u \sin^b(\ell - u)$ and a moving-frame treatment of the kink $|\sin(u-s)|$. The same estimates, used to first order in the other exponent, show that ϕ_k is strictly decreasing for all sufficiently large k. This answers BLN's monotonicity question asymptotically. The conjecture thus remains open only on a finite but unspecified range $440 < k < K_0$. We also isolate an "endpoint monotonicity" conjecture which implies $\phi_{k+1} < \phi_k$.

1. Introduction

Let $\mathbb{S} = \mathbb{R}/2\pi\mathbb{Z}$ and $B_k = \operatorname{conv} U_k(\mathbb{S}) \subset \mathbb{R}^{2k}$, the Barvinok–Novik orbitope [1, 3]. A *raked trigonometric polynomial* of degree $\le 2k-1$ is $f(t) = c + \sum_{j=1}^{k}(a_j \cos(2j-1)t + b_j \sin(2j-1)t)$, i.e. $f = c + \langle C, U_k\rangle$. Faces of B_k through $U(t_1), \dots, U(t_n)$ correspond to nonnegative raked polynomials vanishing at the t_i.

Definition 1 ([2])**.** ϕ_k is the largest number such that for every open arc $\Gamma \subset \mathbb{S}$ of length ϕ_k and all distinct $t_1, \dots, t_n \in \Gamma$, $n \le k$, the set $\operatorname{conv}\{U(t_1), \dots, U(t_n)\}$ is a face of B_k.

Known. Barvinok–Novik [1]: $\phi_k > 0$. Lee: $\phi_k > \sqrt{6}\, k^{-3/2}$. Barvinok–Lee–Novik [2]: $\pi/2 < \phi_k < \pi$ and $\phi_k \to \pi/2$; $\phi_2 = 2\pi/3$, $\phi_3 = \pi - \arccos\frac{3-\sqrt{5}}{2}$, and $\phi_4 \approx 1.8706585$ in closed form. They also give a characterization (Theorem 13 below). Edges of B_k were determined in [1, 4]. The facial structure of B_3 is unknown.

Conjecture 2 ([2])**.** *In Theorem 13(2) one can take $m_a = m_b = k$ if k is even and $\{m_a, m_b\} = \{k-1, k+1\}$ if k is odd. For even k this is equivalent to $\phi_k = 2\alpha_k$, where α_k is the smallest positive root of $\cos\alpha + 1 + \sum_{j=1}^{k-1}(-1)^j \frac{(2j-1)!!}{(2j)!!}\tan^{2j}\alpha = 0$.*

BLN also write that their computations suggest ϕ_k is decreasing, but they could not prove it.

Date: October 27, 2026.

Results. For a *split* (p,q), $p,q\ge 1$, $p+q=k$, and $L\in(0,\pi)$, let $f_{p,q,L}$ be the unique raked polynomial of degree $\le 2k-1$ with constant term 1, a root of multiplicity $2p$ at 0 and of multiplicity $2q$ at L [2, Cor. 3.5]. Let $L(p,q)\in(0,\pi]$ be the first L at which $f_{p,q,L}$ acquires an additional root ($L(p,q)=\pi$ if none). Put $a=2p-1$, $b=2q-1$.

Theorem 3 (integral criterion)**.** *For $L\in(0,\pi)$ let $A(u)=\sin^a u\,\sin^b(L-u)$ on $(0,L)$, let $s\in(0,L)$ be the argument of $\int_0^L A(u)e^{iu}du$, and put*

$$I_{\text{in}}=\int_0^s A(u)\sin(s-u)\,du=\tfrac12\int_0^L A(u)|\sin(u-s)|\,du,\qquad I_{\text{out}}=\int_L^\pi \sin^a u\,\sin^b(u-L)\,\sin(u-s)\,du.$$

Then $f_{p,q,L}\ge 0$ on $\mathbb{S}$ iff $I_{\text{out}}\ge I_{\text{in}}$, and $f_{p,q,L}$ is ≥ 0 and has a root outside $\{0,L\}$ iff $I_{\text{out}}=I_{\text{in}}$. Consequently, $f_{p,q,L}$ has a root outside $\{0,L\}$ iff $I_{\text{out}}\le I_{\text{in}}$. Moreover $\Delta_{p,q}(L):=I_{\text{out}}-I_{\text{in}}$ is positive on $(0,L(p,q))$, vanishes at $L(p,q)$ if $L(p,q)<\pi$, and is negative on $(L(p,q),\pi)$. Finally $\phi_k=\min_{p+q=k}L(p,q)$ and $L(p,q)=L(q,p)$.

Theorem 4 (upper bounds)**.** *(i) For even $k\ge 4$: $\phi_k\le L(\frac{k}{2},\frac{k}{2})<\dfrac{\pi}{2}+\dfrac{\log\left(8\pi(k+\frac12)\right)}{4(k-1)}$.*

(ii) For odd $k\ge 13$: $\phi_k<\dfrac{\pi}{2}+\dfrac{\log\left(64\sqrt{\pi}\,(k-1)\sqrt{k-1/2}\right)}{2k-8}=\dfrac{\pi}{2}+\dfrac{3\log k+O(1)}{4k}$.

Theorem 5 (lower bounds)**.** *(i)* $\liminf_{k\to\infty}\left(4k(\phi_k-\frac{\pi}{2})-\log k\right)\ge 2\log 2$.
(ii) $\phi_k>\frac{\pi}{2}+\frac{\log k}{4k}$ for every $k\ge 2$.

The proof of (ii) is analytic for $k>2\cdot 10^4$. For $301\le k\le 2\cdot 10^4$ it consists of certifying in ball arithmetic, for each of the $\approx 10^8$ splits, the sign of an explicit closed form in Gamma functions (Proposition 25). For $k\le 300$ it follows from the enclosures of Theorem 8.

Theorem 6 (effective upper bound)**.** $\phi_k<\frac{\pi}{2}+\frac{\log k+4.1}{4k}$ *for every $k\ge 2$.*

For even $k\ge 302$ this follows from Theorem 4(i), and for $k\le 300$ from Theorem 8. For odd $k\ge 301$ it comes from an explicit criterion for the split $(\frac{k-1}{2},\frac{k+1}{2})$ (Proposition 19). That criterion is certified in ball arithmetic for $k\le 200001$ and verified by hand beyond. Together with Theorem 5(ii),

$$\frac{\pi}{2}+\frac{\log k}{4k}<\phi_k<\frac{\pi}{2}+\frac{\log k+4.1}{4k}\qquad\text{for every } k\ge 2.$$

Theorem 7 (sharp asymptotics)**.** $\lim\limits_{k\to\infty}\Big(4k\big(\phi_k-\frac{\pi}{2}\big)-\log k\Big)=\log(8\pi)$, *i.e.* $\phi_k=\frac{\pi}{2}+\frac{\log k+\log(8\pi)}{4k}+o\big(\frac{1}{k}\big)$.

This is exactly the asymptotics of the balanced split. It is what Conjecture 2 predicts, but it does not imply the conjecture: a non-balanced split could still win at order $o(1/k)$. The proof (§6) is analytic but does not give an explicit rate. The cruder bounds of Theorems 4, 5(ii) and 6 hold for every k in their stated ranges; Theorem 5(i) is an asymptotic statement.

Theorem 8 (computer-assisted, rigorous)**.** *For every $2\le k\le 440$, the balanced split is the unique minimizer in Theorem 13(2). That is, Conjecture 2 holds, in both parities, and $\phi_k=L(\lfloor k/2\rfloor,\lceil k/2\rceil)$. Moreover $\phi_{k+1}<\phi_k$ for $2\le k<440$, and $L(p,q+1)<L(p,q)$ for all $p+q<200$. Each ϕ_k is enclosed in an interval of half-width $1.01\cdot 10^{-12}$ about a floating-point value, printed to 12 decimals (files `numerics/rigorous_200.txt`, `numerics/rigorous_201_300.txt`, `numerics/rigorous_301_400.txt` and `numerics/rigorous_401_440.txt`).*

BLN proved $k\le 4$. The proof is in §8: Theorem 3 turns each comparison into the sign of an explicit closed-form expression at a rational point, and these signs are certified in ball arithmetic (Arb).

Theorem 9 (all sufficiently large k)**.** *There is K_0 such that for every $k \geq K_0$ the balanced split is the unique minimizer in Theorem 13(2), up to $p \leftrightarrow q$. That is, Conjecture 2 holds for all $k \geq K_0$.*

Theorem 10 (monotonicity for large k)**.** *There is K_2 such that $\phi_{k+1} < \phi_k$ for all $k \geq K_2$. Moreover, endpoint monotonicity $L(p, q+1) < L(p, q)$ (Conjecture 11 below) holds for all $k = p + q \geq K_2$ with $0.42k \leq p \leq 0.58k$.*

The proofs (§7) are analytic but ineffective: we do not compute K_0 or K_2. Together with Theorem 8, the conjecture holds for $k \leq 440$ and for $k \geq K_0$. The splits far from balanced are handled by the sharp asymptotics. For near-balanced splits we prove that the criterion $\log I_{\text{in}} - \log I_{\text{out}}$ is strictly concave in the (real) exponent $2p-1$ at $L = L(\lfloor \frac{k}{2} \rfloor, \lceil \frac{k}{2} \rceil)$. Its second derivative is at most $-4\delta/N + O(N^{-2})$ with $N\delta \asymp \log k$.

In floating point (§9) we also find, for $k \leq 300$, that $p \mapsto L(p, k-p)$ is strictly decreasing on $p \leq k/2$, and that $4k(\phi_k - \pi/2) - \log k$ approaches $\log(8\pi) = 3.2242$, the value of the upper bound in Theorem 4(i). So the upper bound (i) is asymptotically sharp if the conjecture holds. We also propose:

Conjecture 11 (endpoint monotonicity)**.** *$L(p, q+1) < L(p, q)$ for all $p, q \geq 1$.*

Conjecture 11 implies $\phi_{k+1} \leq L(p_*, q_* + 1) < L(p_*, q_*) = \phi_k$, where (p_*, q_*) attains ϕ_k. So it answers BLN's monotonicity question. It is rigorously verified for $p + q < 200$ (Theorem 8).

Relation to the source paper. On an arc of length $< \pi$ the curve U_k meets every hyperplane at most $2k$ times, counting multiplicity [2, Thm. 3.1]; that is, it is a convex curve. The configurations in Theorem 3 are the endpoint principal representations, the analogue in even dimension of the upper and lower hulls in the source paper's Theorem 2.1. ϕ_k is the first time the corresponding supporting hyperplane of the convex hull of the *arc* stops supporting the whole closed curve. The criterion follows the source paper's method: factor the derivative of the support function into explicit sines, and read off signs from integrals of positive functions.

2. Facts from [2]

Theorem 12 ([2], Thm. 3.1, Cor. 3.5)**.** *(a) A nonzero raked f of degree $\leq 2k-1$ has at most $4k-2$ roots on $\mathbb{S}$ counting multiplicity. If its constant term is 0, then $f(t+\pi) = -f(t)$, and it has at most $2k-1$ roots in any open semicircle. (b) If distinct t_i lie in an open semicircle and $\sum m_i = 2k$, there is a unique raked f of degree $\leq 2k-1$ with constant term 1 having a root of multiplicity exactly m_i at t_i.*

Theorem 13 ([2], Thm. 5.1, Lemma 5.7)**.** *(1) For arcs of length $< \phi_k$: if $t_1, \ldots, t_n \in \Gamma$ are distinct and m_i are positive even integers with $\sum m_i = 2k$, then the polynomial of Theorem 12(b) has no other roots and is ≥ 0. ϕ_k is the largest length with this property. (2) For an arc of length ϕ_k with endpoints a, b there are positive even $m_a + m_b = 2k$ such that the polynomial with roots of multiplicities m_a, m_b at a, b is ≥ 0 and has a root in $\Gamma + \pi$. (3) Let Γ be an open arc of length $< \pi$ with endpoint a, and fix positive even m_a, m_b with $m_a + m_b = 2k$. For $b \in \Gamma$ let f_b be the polynomial of Theorem 12(b) with roots of multiplicities m_a at a and m_b at b. Let $x, y, z \in \Gamma$, with x between a and y and y between a and z. If $f_y \geq 0$ and f_y has a root in $\Gamma + \pi$, then $f_x > 0$ off $\{a, x\}$, and f_z is negative somewhere.*

3. The two-cluster criterion: proof of Theorem 3

Lemma 14. *Let g be a raked polynomial of degree $\leq 2k-1$ with zero constant term, i.e. an odd-harmonic trigonometric polynomial. If g has roots of multiplicity $\geq \mu_j$ at points r_j that are distinct mod π, with $\sum \mu_j = 2k-1$, then $g(t) = C \prod_j \sin^{\mu_j}(t - r_j)$.*

Proof. $\cos((2j-1)t)$ and $\sin((2j-1)t)$ are homogeneous polynomials of degree $2j-1$ in $(\cos t, \sin t)$. Multiplying by $(\cos^2 t + \sin^2 t)^{k-j}$, odd-harmonic polynomials of degree $\le 2k-1$ are exactly the restrictions $F(\cos t, \sin t)$ of binary forms F of degree $2k-1$. A root of order μ at $t = r$ means the linear form $\ell_r(x,y) = x\sin r - y\cos r$ divides F to order μ, and $\ell_r(\cos t, \sin t) = -\sin(t-r)$. Distinct r_j mod π give non-proportional linear forms. Hence $\prod \ell_{r_j}^{\mu_j}$, of degree $2k-1$, divides F, and F is a scalar multiple of it. □

Proof of Theorem 3. Write $f = f_{p,q,L}$. Then f' is odd-harmonic of degree $\le 2k-1$, with roots of multiplicity exactly $2p-1$ at 0 and $2q-1$ at L (Theorem 12(b)). Its antipodes are roots as well, since $f'(t+\pi) = -f'(t)$. By Rolle's theorem there is a further root $\sigma \in (0, L)$. Lemma 14 gives $f'(t) = K\, g(t)$ with $g(t) = \sin^a t \sin^b(t-L)\sin(t-\sigma)$. Since $f(0) = f(L) = 0$, $\int_0^L g = 0$. On $(0, L)$, $g = -A(u)\sin(u-\sigma)$ (b is odd), so $\operatorname{Im}\left(e^{-i\sigma}\int_0^L A e^{iu}\right) = 0$. The point $\int_0^L A e^{iu} du / \int_0^L A$ lies in the closed circular segment bounded by the arc $\{e^{iu} : 0 \le u \le L\}$ and its chord. This is a convex set not containing 0, since $L < \pi$. The point is not one of the extreme points 1, e^{iL}, because $A > 0$ on $(0, L)$. The only points of the segment with argument 0 or L are 1 and e^{iL}, so $s \in (0, L)$. The condition above says $\sigma \equiv s \pmod{\pi}$, and since $\sigma, s \in (0, L)$ with $L < \pi$, $\sigma = s$. Put $F(t) = \int_0^t g$. On $(0, s)$, $g > 0$; on (s, L), $g < 0$; on (L, π), $g > 0$, since every factor is positive there ($u - s \in (0, \pi)$). Hence $F > 0$ on $(0, L) \cup (L, \pi]$ and $F(\pi) = \int_L^\pi g = I_{\text{out}} > 0$. Since $g(t+\pi) = -g(t)$, we get $F(t+\pi) = F(\pi) - F(t)$, and F has mean $F(\pi)/2 > 0$. By uniqueness, $f = 2F/F(\pi)$. On $[0, \pi]$, $F \ge 0$ with maximum $\max(F(s), F(\pi))$. On $[\pi, 2\pi]$, $F = F(\pi) - F(\cdot - \pi) \ge 0$ iff $F(s) \le F(\pi)$, with an extra zero exactly when $F(s) = F(\pi)$. If $F(s) > F(\pi)$, then $F(\pi + s) = F(\pi) - F(s) < 0 < F(\pi)$, so F vanishes somewhere in $(\pi, \pi + s)$, which contains neither 0 nor L. Finally $F(s) = I_{\text{in}}$, and $F(\pi) = I_{\text{out}}$.

Sign pattern. Suppose $\Delta(y) = 0$ for some $y \in (0, \pi)$. Then $f_{p,q,y} \ge 0$, and its extra root $s+\pi$ lies in $(\pi, \pi + y)$. Given $0 < x < y < z < \pi$, choose $\gamma \in (z, \pi)$ and apply Theorem 13(3) with $a = 0$, $\Gamma = (0, \gamma)$, $m_a = 2p$, $m_b = 2q$; note $(\pi, \pi + y) \subset \Gamma + \pi$. It gives $\Delta(x) > 0$ and $\Delta(z) < 0$. So Δ has at most one zero, and if $\Delta(y) = 0$ then $\Delta > 0$ before y and $\Delta < 0$ after y. By Theorem 13(1), $\Delta > 0$ for small L. Δ is continuous in L, because A, s and the integrals are. This gives the sign pattern.

$\phi_k = \min L(p, q)$. If $L < \phi_k$, the configuration $\{0, L\}$ lies in an open arc of length $< \phi_k$, so by Theorem 13(1) $\Delta_{p,q}(L) > 0$ for every split, and $L(p, q) \ge \phi_k$. By (2), some split has $\Delta_{p,q}(\phi_k) = 0$.
Symmetry. $t \mapsto f_{p,q,L}(L - t)$ is raked with constant term 1, so it equals $f_{q,p,L}$. □

4. Upper bounds

Proposition 15 (balanced split)**.** *Let k be even, $p = q = k/2$, and $c = \cos(L/2)$. Then*

$$\Delta(L) = c^{2k-1} B(\tfrac{1}{2}, k) - J_k(c), \qquad J_k(c) := \int_c^1 (x^2 - c^2)^{k-1}\, dx.$$

Proof. Here $a = b = k-1$, and A is symmetric about $L/2$, so $s = L/2$. Substituting $u = L/2 - w$, we have $\sin u \sin(L-u) = \cos^2 w - c^2$, and $I_{\text{in}} = \int_0^{L/2} (\cos^2 w - c^2)^{k-1} \sin w\, dw = J_k(c)$. Substituting $u = \frac{L+\pi}{2} + w$, we have $\sin u \sin(u - L) = c^2 - \sin^2 w$ and $\sin(u - s) = \cos w$. Hence $I_{\text{out}} = \int_{-c}^{c} (c^2 - y^2)^{k-1} dy = c^{2k-1} B(\frac{1}{2}, k)$. □

Lemma 16. *$J_m(c) \ge (1-c^2)^m/(2m)$ for $c \in (0, 1)$, $m \ge 1$.*

Proof. With $x = 1 - y$: $x^2 - c^2 \ge (1-c^2) - 2y \ge 0$ for $0 \le y \le (1-c^2)/2 \le 1 - c$. Integrate $((1-c^2) - 2y)^{m-1}$ over $[0, (1-c^2)/2]$. □

Lemma 17 (Wendel [5])**.** *For $x > 0$: $x/\sqrt{x + 1/2} \le \Gamma(x + \frac{1}{2})/\Gamma(x) \le \sqrt{x}$.*

Proof of Theorem 4(i). By Theorem 3 it suffices to show $\Delta(\frac{\pi}{2}+\delta) < 0$ for $\delta = \delta_+ := \log(8\pi(k+\frac{1}{2}))/(4(k-1))$, which satisfies $\delta_+ \le 0.4$ for $k \ge 4$. By Proposition 15 and Lemma 16 it suffices that $c^{2k-1}B(\frac{1}{2},k) < (1-c^2)^k/(2k)$, i.e.

$$2k\log\tan\alpha > \log\big(2kB(\tfrac{1}{2},k)/c\big), \qquad \alpha = \tfrac{\pi}{4}+\tfrac{\delta}{2},\ c=\cos\alpha.$$

We have $\log\tan\alpha = 2\operatorname{artanh}\tan\frac{\delta}{2} \ge \delta$ and $c = (\cos\frac{\delta}{2}-\sin\frac{\delta}{2})/\sqrt{2} \ge (1-\delta)/\sqrt{2}$. By Lemma 17, $B(\frac{1}{2},k) = \sqrt{\pi}\,\Gamma(k)/\Gamma(k+\frac{1}{2}) \le \sqrt{\pi(k+\frac{1}{2})}/k$. So the right side is at most $\frac{1}{2}\log(8\pi(k+\frac{1}{2})) - \log(1-\delta) \le \frac{1}{2}\log(8\pi(k+\frac{1}{2})) + 2\delta$. This is $\le 2k\delta$ for $\delta \ge \delta_+$. Since $\log\tan\alpha > \delta$ strictly for $\delta > 0$, the required strict inequality holds at $\delta = \delta_+$. □

Odd k: a three-cluster configuration. Let k be odd and $\alpha \in (0,\pi/2)$. Consider roots of multiplicity $k-1$ at $\pm\alpha$ and 2 at 0. All multiplicities are even, they sum to $2k$, and they lie in an arc of length 2α. By Theorem 13(1), if the corresponding f is negative somewhere, then $\phi_k \le 2\alpha$.

Proposition 18. *Let $c = \cos\alpha$, $w(y) = (y^2-c^2)^{k-2}$ on $[c,1]$, and $c_\sigma^2 = \int_c^1 wy^2/\int_c^1 w$. Then $f \ge 0$ on $\mathbb{S}$ iff*

$$\text{LHS} := \int_{c_\sigma}^1 w(y)(y^2-c_\sigma^2)\,dy \ \le\ \text{RHS} := 2\int_0^c (c^2-y^2)^{k-2}(c_\sigma^2-y^2)\,dy.$$

Proof. By uniqueness f is even. f' is odd-harmonic of degree $\le 2k-1$, with roots of multiplicity $k-2$ at $\pm\alpha$ and 1 at 0, so it is divisible by $(\sin^2 t - \sin^2\alpha)^{k-2}\sin t$ in the binary-form sense. The quotient Q is a quadratic form, i.e. $Q = A + B\cos 2t + C\sin 2t$, and it is even, so $C = 0$. Rolle's theorem on $(0,\alpha)$ gives a root σ, so $Q \propto \sin^2 t - \sin^2\sigma$. With $y = \cos u$: $f(t) = K\int_{\cos t}^1 g$, where $g(y) = (c^2-y^2)^{k-2}(c_\sigma^2-y^2)$. The condition $f(\alpha) = 0$ is $\int_c^1 g = 0$, which gives the formula for c_σ (recall $k-2$ is odd). The sign of g is $+$ on $|y| < c$, $-$ on $c < |y| < c_\sigma$, and $+$ on $|y| > c_\sigma$. Also $f(\pi) = 2K\int_0^c g > 0$ equals twice the mean, so $K > 0$. Tracking $\int_x^1 g$ for x from 1 down to -1, its only possible negative value is at $x = -c_\sigma$, where it equals $2\int_0^c g - \int_{c_\sigma}^1 g$, and $\int_{c_\sigma}^1 g = \text{LHS}$. By evenness this settles $[-\pi,0]$ as well. □

Proof of Theorem 4(ii). First, RHS $\le c^{2k-3}B(\frac{1}{2},k-1)$. For LHS: let $Y = y^2$ with $y \sim w/Z$ on $[c,1]$, $Z = J_{k-1}(c)$. Then LHS $= Z\,\mathbb{E}(Y-\mathbb{E}Y)_+ = \frac{Z}{2}\mathbb{E}|Y-\mathbb{E}Y|$. For any density with maximum $f_{\max}$, $\mathbb{E}|Y-\mathbb{E}Y| \ge 1/(4f_{\max})$, since $\mathbb{P}(|Y-\mathbb{E}Y| < r) \le 2rf_{\max}$. Here $f_Y(v) = w(\sqrt{v})/(2\sqrt{v}Z) \le (1-c^2)^{k-2}/(2cZ)$. With Lemma 16 this gives LHS $\ge cZ^2/(4(1-c^2)^{k-2}) \ge c(1-c^2)^k/(16(k-1)^2)$. So f is negative somewhere if $2k\log\tan\alpha > \log\big(16(k-1)^2B(\frac{1}{2},k-1)\big) - 4\log c$. Put $\alpha = \frac{\pi}{4}+\frac{\delta}{2}$ and use $\log\tan\alpha \ge \delta$, $-4\log c \le 2\log 2 + 8\delta$ for $\delta \le 1/2$, and $B(\frac{1}{2},k-1) \le \sqrt{\pi(k-\frac{1}{2})}/(k-1)$. It suffices that $(2k-8)\delta > \log(64\sqrt{\pi}(k-1)\sqrt{k-1/2})$. The resulting δ is $\le 1/2$ for $k \ge 13$. Strictness: put $\delta_0 = \log(64\sqrt{\pi}(k-1)\sqrt{k-1/2})/(2k-8)$. At $\delta = \delta_0$ the sufficient inequality still holds strictly, because $\log\tan\alpha > \delta$ for $\delta > 0$. So f is negative somewhere at $\alpha_0 = \frac{\pi}{4}+\frac{\delta_0}{2}$. The coefficients of f solve a linear system (constant term 1, and roots of multiplicity at least $k-1, 2, k-1$ at $-\alpha, 0, \alpha$) whose matrix is continuous in α. It is invertible: a kernel vector would be a nonzero raked polynomial with zero constant term and at least $2k$ roots in the open semicircle $(-\frac{\pi}{2},\frac{\pi}{2})$, which Theorem 12(a) excludes. So f depends continuously on α, and negativity is an open condition in α. Hence f is negative somewhere at some $\alpha' < \alpha_0$, and $\phi_k \le 2\alpha' < \frac{\pi}{2}+\delta_0$. □

Odd k: the sharp order of magnitude. For odd k the three-cluster bound loses a factor 3 in front of $\log k$. The near-balanced split $p = \frac{k-1}{2}$, $q = \frac{k+1}{2}$ recovers the factor 1. The tools are those of §5 below, used in the opposite direction.

Proposition 19 (general-split upper criterion)**.** *Let $2 \le p < q$ with $b - a = 2$, and put $N = a + b$, $L = \frac{\pi}{2} + \delta$, $t = \tan\delta$, $w = N^{-1/3}$. Assume $\frac{L}{2} - \frac{2}{a} - w > 0$, $\frac{L}{2} + \frac{2}{a} + w \le \frac{\pi}{2}$, and $\kappa := (\sqrt{a} + \sqrt{b})^2/(1 + \sin\delta) > 4$. Put*

$$K_w = \frac{a}{\sin^2(\frac{L}{2} - \frac{2}{a} - w)} + \frac{b}{\sin^2(\frac{L}{2} - w)}, \qquad E_{\mathrm{low}} = \tfrac{1}{4}\sqrt{2\pi/K_w}\,\big(1 - \mathrm{erfc}(w\sqrt{K_w/2})\big) - \tfrac{1}{6}(152/\kappa^2)^{3/4}.$$

Let $D_1 = 2\Gamma(p+\frac{1}{2})\Gamma(q+\frac{1}{2})/(\Gamma(p)\Gamma(q))$ as in Lemma 24(c). If $E_{\mathrm{low}} > 0$ and $(\delta+t)D_1 + b\log\cos\delta - \frac{bt^2 p}{2(q-1)} > \log 2 - \log E_{\mathrm{low}}$, then $\Delta_{p,q}(L) < 0$, and hence $\phi_k \le L(p,q) < L$.

Proof. We use the notation $M_{x,y}(\ell) = \int_0^\ell \sin^x u \, \sin^y(\ell - u)\, du$ and Lemmas 22 and 24 of §5 below; their proofs do not use this subsection. *Outer integral.* $I_{\mathrm{out}} \le M_{a,b}(\pi - L)$ since $\sin \le 1$, and by log-concavity (Lemma 24(a),(c)) $\log M_{a,b}(\frac{\pi}{2} - \delta) \le \log M_{a,b}(\frac{\pi}{2}) - \delta D_1$.

Inner mass. Dropping $u \in (\frac{\pi}{2}, L)$, we get $M_{a,b}(L) \ge \cos^b\delta\, M_{a,b}(\frac{\pi}{2})\, \mathbb{E}_0(1+tT)^b$, where $u \sim \sin^a u \cos^b u$ on $(0, \frac{\pi}{2})$ and $T = \tan u$. By Jensen and $\log(1+x) \ge x - x^2/2$, $\mathbb{E}_0(1+tT)^b \ge \exp(tb\mathbb{E}_0 T - \frac{bt^2}{2}\mathbb{E}_0 T^2)$, with $b\mathbb{E}_0 T = bB(p+\frac{1}{2}, q-\frac{1}{2})/B(p,q) = D_1$ (as in the proof of Lemma 24(c)) and $\mathbb{E}_0 T^2 = B(p+1, q-1)/B(p,q) = p/(q-1)$.

Mean absolute deviation. $I_{\mathrm{in}} = \frac{1}{2} M_{a,b}(L)\, \mathbb{E}|\sin(u-s)|$ for $u \sim A/M$ on $(0, L)$, $A = e^{-V}$. Let u_1 be the mode, so $V'(u_1) = 0$. Since $V'(\frac{L}{2}) = (b-a)\cot\frac{L}{2} > 0$ and $V'(\frac{L}{2} - \eta) \le -2a\eta + (b-a)$ (because $\cot(x-\eta) - \cot(x+\eta) \ge 2\eta$ and $\cot(\frac{L}{2} + \eta) \le 1$), we have $u_1 \in (\frac{L}{2} - \frac{2}{a}, \frac{L}{2})$. On $|u - u_1| \le w$ we have $V'' \le K_w$ by the two assumptions on the window. Hence $A(u) \ge A(u_1) e^{-K_w(u-u_1)^2/2}$ there, and the maximal density satisfies $A(u_1)/M \le 1/\int_{-w}^{w} e^{-K_w x^2/2} dx$. Since $\mathbb{P}(|u-s| < r) \le 2rA(u_1)/M$, $\mathbb{E}|u-s| \ge M/(4A(u_1))$. Integrating $(u-u_1)^{2j-1}V'(u)$ by parts, with $V'' \ge \kappa$ (proof of Lemma 22), gives $\mathbb{E}(u-u_1)^2 \le 1/\kappa$ and $\mathbb{E}(u-u_1)^4 \le 3/\kappa^2$. Hence $\mathbb{E}\cos(u-u_1) \ge 1 - \frac{1}{2\kappa} \ge \frac{1}{2}$ and $|\mathbb{E}\sin(u-u_1)| \le \mathbb{E}|u-u_1| \le \kappa^{-1/2} < \frac{1}{2}$, since $\kappa > 4$. The angle s maximizes $\sigma \mapsto \mathbb{E}\cos(u-\sigma)$, so $\mathbb{E}\cos(u-s) \ge \mathbb{E}\cos(u-u_1)$. If $|s-u_1| \ge \frac{\pi}{2}$, then $\mathbb{E}\cos(u-s) = \cos(s-u_1)\mathbb{E}\cos(u-u_1) + \sin(s-u_1)\mathbb{E}\sin(u-u_1) \le |\mathbb{E}\sin(u-u_1)| < \frac{1}{2} \le \mathbb{E}\cos(u-u_1)$, a contradiction. So $|s-u_1| < \frac{\pi}{2}$. From $\mathbb{E}\sin(u-s) = 0$ we get $\tan(s-u_1) = \mathbb{E}\sin(u-u_1)/\mathbb{E}\cos(u-u_1)$, hence $|s-u_1| \le |\tan(s-u_1)| \le \kappa^{-1/2}/(1-\frac{1}{2\kappa}) \le 2\kappa^{-1/2}$. Thus $\mathbb{E}(u-s)^4 \le 8(3+16)/\kappa^2$. Finally $\mathbb{E}|\sin(u-s)| \ge \mathbb{E}|u-s| - \frac{1}{6}\mathbb{E}|u-s|^3 \ge E_{\mathrm{low}}$. Combining, $\log I_{\mathrm{in}} - \log I_{\mathrm{out}} \ge (\delta+t)D_1 + b\log\cos\delta - \frac{bt^2 p}{2(q-1)} - \log 2 + \log E_{\mathrm{low}} > 0$. □

Proof of Theorem 6. Put $C = 4.1$ and $\delta_k = (\log k + C)/(4k)$.

$2 \le k \le 300$. The enclosures of Theorem 8 give $\phi_k < \frac{\pi}{2} + \delta_k$, compared in ball arithmetic. The smallest margin is $7.2 \cdot 10^{-4}$, at $k = 300$ (`numerics/bln_odd_upper_arb.py`, output `numerics/odd_upper_arb.txt`).

Even $k \ge 302$. By Theorem 4(i), $4k(\phi_k - \frac{\pi}{2}) - \log k < g(k) := \log(8\pi) + \log(1+\frac{1}{2k}) + \frac{\log(8\pi(k+1/2))}{k-1}$. Both k-dependent terms decrease in k; for the last one,

$$\frac{d}{dx}\,\frac{\log(8\pi(x+\frac{1}{2}))}{x-1} = \frac{(x-1)/(x+\frac{1}{2}) - \log(8\pi(x+\frac{1}{2}))}{(x-1)^2} < 0.$$

Hence $g(k) \le g(302) < 3.26 < C$.

Odd $301 \le k \le 200001$. Apply Proposition 19 with $p = \frac{k-1}{2}$, $q = \frac{k+1}{2}$ (so $a = k-2$, $b = k$) and $\delta = \delta_k$. For each of these 99851 values of k, its four hypotheses and its final inequality were certified in Arb ball arithmetic (precision 96 bits). The smallest certified margin in the final inequality is 0.0257, at $k = 301$ (same script).

Odd $k \ge 200003$. We verify the hypotheses and the inequality of Proposition 19 by hand. Let $N = 2k - 2$ and $w = N^{-1/3}$, and put $\varepsilon_1 = \frac{2}{a} + w$. Then $\delta := \delta_k \le 2.04 \cdot 10^{-5}$ and $\varepsilon_1 \le 0.0135821$, and both decrease in k. So $\frac{L}{2} - \frac{2}{a} - w = \frac{\pi}{4} + \frac{\delta}{2} - \varepsilon_1 > 0$ and $\frac{L}{2} + \frac{2}{a} + w < \frac{\pi}{2}$;

and $\kappa \geq 4a/(1+\delta) > 4$, since $b > a$. Both arguments of sin in K_w lie in $[\frac{\pi}{4} - \varepsilon_1, \frac{\pi}{2}]$. Moreover $\sin(\frac{\pi}{4} - \varepsilon) = (\cos\varepsilon - \sin\varepsilon)/\sqrt{2} \geq s(\varepsilon)/\sqrt{2}$, where $s(\varepsilon) = 1 - \varepsilon - \varepsilon^2/2$. Hence $K_w \leq 2N/s(\varepsilon_1)^2$, and $\frac{1}{4}\sqrt{2\pi/K_w} \geq \frac{1}{4}\sqrt{\pi/N}\, s(\varepsilon_1)$. Since $\sin^2 \leq 1$, $K_w \geq a + b = N$, so $\operatorname{erfc}(w\sqrt{K_w/2}) \leq e_1 := \operatorname{erfc}(N^{1/6}/\sqrt{2}) < 10^{-16}$. From $\kappa \geq 4a/(1+\delta)$ we get $\frac{1}{6}(152/\kappa^2)^{3/4} \leq 0.902\, a^{-3/2} \leq \frac{1}{4}\sqrt{\pi/N} \cdot \frac{2.9}{a}$, using $N = 2a + 2$. Therefore

$$E_{\text{low}} \geq \tfrac{1}{4}\sqrt{\pi/N}\,(1-\eta), \qquad \eta := \varepsilon_1 + \tfrac{1}{2}\varepsilon_1^2 + e_1 + \tfrac{2.9}{a} \leq 0.01369.$$

In particular $E_{\text{low}} > 0$, and since $N < 2k$, $\log 2 - \log E_{\text{low}} < \frac{1}{2}\log k + \frac{1}{2}\log\frac{128}{\pi} - \log(1-\eta)$. On the other side, Lemma 17 gives $D_1 \geq 2pq/\sqrt{(p+\frac{1}{2})(q+\frac{1}{2})} = (k^2-1)/\sqrt{k(k+2)} \geq k - 1$. Since $\delta \leq t \leq 2\delta$, $\log\cos\delta \geq -\delta^2$ and $q - 1 = p$, the left side of the inequality in Proposition 19 is at least $2\delta(k-1) - k\delta^2 - \frac{k}{2}t^2 \geq \frac{1}{2}(\log k + C) - 2\delta - 3k\delta^2$. The quantity $3k\delta^2 = 3(\log k + C)^2/(16k)$ decreases in k and is $\leq 2.5 \cdot 10^{-4}$. So the margin in the inequality of Proposition 19 is at least

$$\tfrac{C}{2} - \tfrac{1}{2}\log\tfrac{128}{\pi} + \log(1-\eta) - 2\delta - 3k\delta^2 \;\geq\; 2.05 - 1.85366 - 0.01379 - 0.00005 - 0.00025 \;>\; 0.18.$$

In all cases Proposition 19 or the preceding cases give $\phi_k < \frac{\pi}{2} + \delta_k$. □

Remark 20. Letting $C \downarrow \log(128/\pi)$ in the last step shows that Proposition 19 alone gives $\limsup_{k \text{ odd}} \big(4k(\phi_k - \frac{\pi}{2}) - \log k\big) \leq \log(128/\pi) = 3.707$. Theorem 7 gives the sharp value $\log(8\pi) = 3.224$ for both parities. The explicit lower bounds on the margin used in the last step are compared with the exact criterion at $k = 200003$, $10^6 + 1$, $10^8 + 1$ and $10^{12} + 1$ in the same script.

5. Lower bound: proof of Theorem 5

Throughout, $p \leq q$ (by symmetry), $a = 2p - 1$, $b = 2q - 1$, $L = \frac{\pi}{2} + \delta$ with $0 < \delta \leq 0.2$, $t = \tan\delta$, and $M_{x,y}(\ell) = \int_0^\ell \sin^x u\, \sin^y(\ell - u)\, du$.

Lemma 21 (variance)**.** *Let $\mu = e^{-V}du$ be a probability measure on an interval, $V \in C^2$, $V'' \geq \kappa > 0$, and $(u-m)e^{-V} \to 0$ at the endpoints, where $m = \int u\, d\mu$. Then $\operatorname{Var}_\mu(u) \leq 1/\kappa$. If moreover the interval has length $< \pi$ and $s = \arg\int e^{iu} d\mu$, then $\int \sin^2(u-s)\, d\mu \leq \operatorname{Var}_\mu(u)$.*

Proof. Integrating by parts, $\int (u-m)V'(u)\, d\mu = 1$. Also $\int (u-m)V'(m)\, d\mu = 0$, and $(u-m)(V'(u) - V'(m)) \geq \kappa(u-m)^2$. Hence $\kappa \operatorname{Var} \leq 1$. Next, $\sigma \mapsto \int \cos(u-\sigma) d\mu = |\int e^{iu} d\mu| \cos(\sigma - s)$ is maximal at $\sigma = s$. Hence $\int \sin^2(u-s) \leq 2\int(1 - \cos(u-s)) \leq 2\int(1 - \cos(u-m)) \leq \int (u-m)^2$. □

Lemma 22. $I_{\text{in}} \leq \frac{1}{2} M_{a,b}(L)\, \kappa_\delta^{-1/2}$ *with* $\kappa_\delta = (\sqrt{a} + \sqrt{b})^2/(1 + \sin\delta)$.

Proof. Apply Lemma 21 to $A/M_{a,b}(L) = e^{-V}$, with $V'' = a/\sin^2 u + b/\sin^2(L-u)$. By Cauchy–Schwarz, V'' is at least $(\sqrt{a} + \sqrt{b})^2/(\sin^2 u + \sin^2(L-u))$, and $\sin^2 u + \sin^2(L-u) = 1 - \cos L \cos(2u - L) \leq 1 + \sin\delta$. Then $I_{\text{in}} = \frac{1}{2} M \int |\sin(u-s)| \leq \frac{1}{2} M (\int \sin^2(u-s))^{1/2}$. □

Lemma 23. $I_{\text{out}} \geq M_{a+1,b+1}(\pi - L)$.

Proof. For $u \in [L, \pi]$ we have $u - s \in [u - L, u] \subset [0, \pi]$, so by concavity of sin, $\sin(u-s) \geq \min(\sin u, \sin(u-L)) \geq \sin u \sin(u-L)$. Substitute $v = u - L$ and use $\sin(L+v) = \sin(\pi - L - v)$. □

Lemma 24. *(a) $\ell \mapsto M_{x,y}(\ell)$ is log-concave on $(0,\pi)$ for $x, y \geq 0$. (b) $M_{a,b}(\pi/2) = \frac{1}{2}B(p,q)$, $M_{a+1,b+1}(\pi/2) = \frac{1}{2}B(p+\frac{1}{2}, q+\frac{1}{2})$. (c) $D_1 := (\log M_{a,b})'(\pi/2) = 2\frac{\Gamma(p+1/2)\Gamma(q+1/2)}{\Gamma(p)\Gamma(q)} \leq 2\sqrt{pq}$. (d) Let $u \sim \sin^{a+1} u \cos^{b+1} u\, du$ on $(0, \pi/2)$ and $T = \tan u$. Then $(b+1)\mathbb{E}T = D_2 := 2\frac{\Gamma(p+1)\Gamma(q+1)}{\Gamma(p+1/2)\Gamma(q+1/2)}$, and $\sin^2 u \sim \operatorname{Beta}(p+\frac{1}{2}, q+\frac{1}{2})$.*

Proof. (a) $M_{x,y} = h_x * h_y$ with $h_x = \sin^x \cdot 1_{(0,\pi)}$ log-concave, and a convolution of log-concave functions is log-concave (Prékopa's theorem [6]). (b),(d) are Beta integrals: $\int_0^{\pi/2} \sin^x u \cos^y u\, du = \frac{1}{2}B(\frac{x+1}{2}, \frac{y+1}{2})$. (c) $M'_{a,b}(\ell) = b\int \sin^a u \sin^{b-1}(\ell-u)\cos(\ell-u)$, which at $\ell = \pi/2$ gives $bB(p+\frac{1}{2}, q-\frac{1}{2})/B(p,q)$; simplify, and use Lemma 17. □

Proposition 25 (explicit safety criterion)**.** *Let* $\tau = (4/5)^{p-1/2}(1/5)^{q+1/2}/B(p+\frac{1}{2}, q+\frac{1}{2})$ *and suppose* $\tau < 1$, $t < 1/2$. *Define*

$$\Phi(p,q,\delta) = \log \tfrac{B(p,q)}{B(p+\frac{1}{2},q+\frac{1}{2})} + \delta D_1 + \frac{tD_2}{(1-2t)(1-\tau)} - (b+1)\log\cos\delta - \log(1-\tau) - \log 2 - \tfrac{1}{2}\log\kappa_\delta.$$

If $\Phi(p,q,\delta) < 0$, *then* $\Delta_{p,q}(\frac{\pi}{2}+\delta) > 0$, *and hence* $L(p,q) > \frac{\pi}{2}+\delta$.

Proof. By Lemmas 22 and 23 it suffices that $\log M_{a,b}(\frac{\pi}{2}+\delta) - \log M_{a+1,b+1}(\frac{\pi}{2}-\delta) < \log 2 + \frac{1}{2}\log\kappa_\delta$. By Lemma 24(a), $\log M_{a,b}(\frac{\pi}{2}+\delta) \le \log M_{a,b}(\frac{\pi}{2}) + \delta D_1$. Next, $M_{a+1,b+1}(\frac{\pi}{2}-\delta) = \int_0^{\pi/2-\delta} \sin^{a+1} u \cos^{b+1}(u+\delta)du$, and $\cos(u+\delta) = \cos\delta\cos u(1-tT)$. So $M_{a+1,b+1}(\frac{\pi}{2}-\delta) = \cos^{b+1}\delta\, M_{a+1,b+1}(\frac{\pi}{2})\,\mathbb{E}(1-tT)_+^{b+1}$. On $\{T \le 2\}$, $\log(1-tT) \ge -tT/(1-2t)$. By Jensen on the conditional law, $\mathbb{E}(1-tT)_+^{b+1} \ge P_0 \exp(-(b+1)t\mathbb{E}T/((1-2t)P_0))$ with $P_0 = \mathbb{P}(T \le 2)$. Moreover $1 - P_0 = \mathbb{P}(\sin^2 u > 4/5) \le \tau$: the Beta$(p+\frac{1}{2}, q+\frac{1}{2})$ density is decreasing on $[4/5, 1]$ because $p \le q$. Combine with Lemma 24(b),(d). □

Proof of Theorem 5(i). Fix $c < 2\log 2$ and put $\delta = (\log k + c)/(4k)$. We show $\max_{p\le q}\Phi < 0$ for large k. Lemma 17 gives $\log\frac{B(p,q)}{B(p+1/2,q+1/2)} \le \log k - \frac{1}{2}\log(pq) + \frac{1}{2}\log\big((1+\frac{1}{2p})(1+\frac{1}{2q})\big)$, $D_2 \le 2\sqrt{(p+\frac{1}{2})(q+\frac{1}{2})}$, and $D_1 \le 2\sqrt{pq}$. Also $\tau \le C\sqrt{k}\,(0.8)^k$ uniformly in $p \le q$, using $1/B(\alpha,\beta) \le 2^{\alpha+\beta-1}\sqrt{(\alpha+\beta)/(2\pi)}$ (log-convexity of Γ and duplication). Further $-\log\cos\delta \le \delta^2$, and $\frac{t}{\delta(1-2t)(1-\tau)} = 1 + O(\delta+\tau)$. Finally $\delta\sqrt{pq}\,(\frac{1}{p}) \le \delta\sqrt{k} \to 0$, $k\delta^2 \to 0$, and $\delta k\,(\delta+\tau) \to 0$. So, uniformly in p,

$$\Phi \le F + o(1), \qquad F = -\log y + (\log k + c)y - \log 2 - \log(\sqrt{a}+\sqrt{b}) + \tfrac{1}{2}\log(1+\tfrac{1}{2p}), \qquad y = \tfrac{\sqrt{pq}}{k} \in (0, \tfrac{1}{2}].$$

Since $(\sqrt{a}+\sqrt{b})^2 = 2k - 2 + 2\sqrt{(2p-1)(2q-1)} \ge 2k(1+2y) - 2 - 2/y$ and $1/y \le k/\sqrt{k-1}$, we have $\log(\sqrt{a}+\sqrt{b}) \ge \frac{1}{2}\log(2k) + \frac{1}{2}\log(1+2y) - o(1)$. Thus $F \le h(y) + \frac{1}{2}\log(1+\frac{1}{2p}) + o(1)$, where $h(y) = -\log y + (\log k + c)y - \frac{1}{2}\log k - \frac{3}{2}\log 2 - \frac{1}{2}\log(1+2y)$ is convex. We have $h(\frac{1}{2}) = \frac{c}{2} - \log 2 < 0$, $h(y_{\min}) = -\frac{3}{2}\log 2 + o(1)$ with $y_{\min} = \sqrt{k-1}/k$, and $h(k^{-1/4}) = -\frac{1}{4}\log k - \frac{3}{2}\log 2 + o(1)$. On $[k^{-1/4}, \frac{1}{2}]$, $p \ge ky^2 \ge \sqrt{k}$, so the extra term is $o(1)$, and by convexity $h \le \max(h(k^{-1/4}), h(\frac{1}{2})) < 0$. On $[y_{\min}, k^{-1/4}]$, the extra term is at most $\frac{1}{2}\log\frac{3}{2} < 0.203$ and $h \le -\frac{3}{2}\log 2 + o(1) < -1.03$. Hence $\Phi < 0$ for all splits when k is large. Proposition 25 and Theorem 3 give $\phi_k > \frac{\pi}{2} + \delta$ for large k. □

Proof of Theorem 5(ii). Take $c = 0$. For $k > 2 \cdot 10^4$ the error terms in the previous proof are explicitly small. Here $\delta \le 1.3 \cdot 10^{-4}$, $(1+3\delta)\delta\sqrt{k}/2 \le 0.009$ (the $1/p$ part of the D_2 correction, including the factor $\frac{t}{(1-2t)(1-\tau)} \le (1+3\delta)\delta$), $3k\delta^2 \le 10^{-3}$ (the $O(\delta)$ part), $2k\delta^2 = (\log k)^2/(8k) \le 7 \cdot 10^{-4}$ (the $\log\cos\delta$ term), the $\log(\sqrt{a}+\sqrt{b})$ correction is $\le (1 + k/\sqrt{k-1})/k < 0.008$, and $\tau < 10^{-900}$. The remaining terms are also small: $\frac{1}{2}\log(1+\sin\delta) \le \delta/2$; $\frac{1}{2}\log(1+\frac{1}{2q}) \le \frac{1}{2k}$ (as $q \ge k/2$); the $1/q$ and $1/(pq)$ parts of the D_2 correction, at most $\frac{t}{(1-2t)(1-\tau)} \cdot 2\sqrt{pq}\,(\frac{1}{4q} + \frac{1}{16pq}) \le \delta$; and $-\log(1-\tau) \le 2\tau$. Together they contribute $< 3 \cdot 10^{-4}$. All these bounds decrease in k, and their total is < 0.02. On the other side, $h(\frac{1}{2}) = -\log 2$, $h(k^{-1/4}) \le -2.7$, and on the small-p branch $h \le h(y_{\min}) \le -0.976$, so even after adding $\frac{1}{2}\log\frac{3}{2}$ it stays ≤ -0.77. On the large-p branch, $p \ge \sqrt{k}$ and the extra term $\frac{1}{2}\log(1+\frac{1}{2p})$ is $\le \frac{1}{4\sqrt{k}} < 0.002$. So in every case the error total is < 0.025, against a margin of at least 0.69. Hence $\Phi < 0$.

For $301 \le k \le 2 \cdot 10^4$ we certified $\Phi(p, k-p, \frac{\log k}{4k}) < 0$, together with the hypotheses $\tau < 1$ and $t < \frac{1}{2}$ of Proposition 25, in Arb ball arithmetic (precision 96 bits) for each of the 99977500 pairs (k, p) with $p \le k/2$. All Gamma arguments are multiples of $\frac{1}{2}$, so $\log\Gamma(n/2)$ is tabulated once. The largest upper endpoint of the certified enclosures of Φ is -0.6127, at $k = 301$, $p = 1$ (`numerics/bln_lower_arb.py`, output `numerics/lower_arb.txt`; about 80 seconds on 8 cores). For $2 \le k \le 300$, the enclosures of Theorem 8 give $\phi_k - \frac{\pi}{2} - \frac{\log k}{4k} > 0$; the smallest margin is 0.0026, at $k = 300$ (same script). □

6. Sharp asymptotics: proof of Theorem 7

Lemma 26 (Laplace bounds for strongly log-concave laws)**.** *Let $\mu = e^{-V}du/Z$ be a probability measure on an open interval J, $V \in C^2(J)$, $V'' \ge \kappa > 0$, and $e^{-V} \to 0$ at the endpoints. Let $V'(u_1) = 0$, $K = V''(u_1)$, $y = u - u_1$. Suppose $|V'' - K| \le \eta K$ on $[u_1 - w, u_1 + w] \subset J$ with $\eta \le \frac{1}{2}$, and put $\lambda = (1-\eta)Kw^2/2 \ge 1$. Then, with an absolute constant C:*

$$\left|\mathbb{E}|y| - \sqrt{2/(\pi K)}\right| \le C(\eta + e^{-\lambda})K^{-1/2}, \qquad |\mathbb{E}y| \le C(\eta+e^{-\lambda})K^{-1/2}, \qquad \mathbb{E}y^2 \le \kappa^{-1}, \quad \mathbb{E}y^4 \le 3\kappa^{-2}.$$

Proof. Let $\varphi(y) = V(u_1 + y) - V(u_1)$ and $K_\pm = (1 \pm \eta)K$. For $|y| \le w$, Taylor's formula gives $K_- y^2/2 \le \varphi \le K_+ y^2/2$. For $|y| > w$, convexity and $\varphi'(\pm w) \gtrless \pm K_- w$ give $\varphi(y) \ge K_- w(|y| - w/2)$. Hence the parts of $\int e^{-\varphi}$ and $\int |y| e^{-\varphi}$ over $|y| > w$ are at most $\frac{2}{K_- w} e^{-\lambda}$ and $\frac{2e^{-\lambda}}{K_- w}(w + \frac{1}{K_- w})$. Over $|y| \le w$ they lie between the corresponding Gaussian integrals with K_+ and K_-, namely $\sqrt{2\pi/K_\pm}\,(1 - O(e^{-\lambda}))$ and $\frac{2}{K_\pm}(1 - O(e^{-\lambda}))$. Taking ratios gives the first bound. For the second, compare $\int_0^w y e^{-\varphi}$ and $\int_{-w}^0 |y| e^{-\varphi}$; both lie in $[\frac{1}{K_+}(1 - e^{-\lambda}), \frac{1}{K_-}]$. The last two bounds come from integrating $y^{2j-1}V'(u_1 + y) \ge \kappa y^{2j}$ by parts ($j = 1, 2$). □

Fix $\epsilon = \frac{1}{5}$ and call a split *central* if $\epsilon k \le p \le q$ and *extreme* if $p < \epsilon k$ (with $p \le q$). Throughout, $o(1)$ is uniform over central splits as $k \to \infty$, and $0 \le \delta \le 2\log k/k$, $L = \frac{\pi}{2} + \delta$, $L' = \pi - L$. Write $u_1(\ell)$ for the mode of $A_\ell = \sin^a u \sin^b(\ell - u)$ on $(0, \ell)$, and $r_{ab} = 2\sqrt{ab}/(a+b)$.

Lemma 27 (central splits)**.** *(a) $\tan^2 u_1(\frac{\pi}{2}) = a/b$, $|u_1'(\ell)| \le 1$, and $K(\ell) := V_\ell''(u_1(\ell)) = 2N(1 + O(|\ell - \frac{\pi}{2}|))$. Moreover $|V_\ell'''| = O(N)$ within distance 0.1 of $u_1(\ell)$. (b) $I_{\mathrm{in}} = \frac{1}{2}M_{a,b}(L)(\pi N)^{-1/2}(1 + o(1))$. (c) $I_{\mathrm{out}} = M_{a,b}(L')\, r_{ab}(1 + o(1))$. (d) $\Psi := \log M_{a,b}(L) - \log M_{a,b}(L') = 2\delta D_1 + O(N\delta^2) + o(1)$, and $D_1 = 2\sqrt{pq}(1 + O(1/k))$.*

Proof. (a) u_1 solves $F(u, \ell) := a\cot u - b\cot(\ell - u) = 0$, and $u_1' = -\partial_\ell F/\partial_u F = \frac{b\csc^2(\ell-u)}{a\csc^2 u + b\csc^2(\ell-u)} \in (0, 1)$. At $\ell = \frac{\pi}{2}$, $\sin^2 u_1 = a/N$, so $K = a \cdot \frac{N}{a} + b \cdot \frac{N}{b} = 2N$. For central splits $\tan^2 u_1(\frac{\pi}{2}) \in [\frac{1}{8}, 1]$, so u_1 and $\ell - u_1$ stay at distance ≥ 0.25 from 0 for $|\ell - \frac{\pi}{2}| \le 0.05$. On that range $\csc^2$ and $\csc^2 \cdot \cot$ are bounded, which gives the remaining claims. (b) Apply Lemma 26 to A_L/M, with $\kappa \ge N/2$ (Lemma 22), $w = N^{-1/3}$, $\eta = O(N^{-1/3})$ and $\lambda \asymp N^{1/3}$. We get $\mathbb{E}|y| = (\pi N)^{-1/2}(1 + o(1))$ and $\mathbb{E}y = o(N^{-1/2})$. From $\mathbb{E}\sin(u - s) = 0$, $\tan(s - u_1) = \mathbb{E}\sin y/\mathbb{E}\cos y$, and $|\mathbb{E}\sin y - \mathbb{E}y| \le \mathbb{E}|y|^3 = O(N^{-3/2})$. To pass to $s - u_1$ itself, note that $\mathbb{E}\cos y \ge 1 - \frac{1}{2}\mathbb{E}y^2 \ge 1 - \frac{1}{N}$. Also s maximizes $\sigma \mapsto \mathbb{E}\cos(u - \sigma)$, so $\mathbb{E}\cos(u - s) \ge \mathbb{E}\cos y$. If $|s - u_1| \ge \frac{\pi}{2}$, then $\mathbb{E}\cos(u - s) \le |\mathbb{E}\sin y| \le \mathbb{E}|y| < \mathbb{E}\cos y$, a contradiction. Hence $|s - u_1| < \frac{\pi}{2}$, and $s - u_1 = \arctan(\mathbb{E}\sin y/\mathbb{E}\cos y) = o(N^{-1/2})$. Hence $\mathbb{E}|\sin(u - s)| = \mathbb{E}|y| + O(|s - u_1| + \mathbb{E}|y|^3 + |s - u_1|^3) = (\pi N)^{-1/2}(1 + o(1))$, and $I_{\mathrm{in}} = \frac{1}{2}M\,\mathbb{E}|\sin(u - s)|$. (c) The substitution $v = \pi - u$ maps the outer density to $A_{L'}$ and $\sin(u - s)$ to $\sin(v + s)$. By (a) and Lemma 26, $\mathbb{E}|v - u_1(L')| = O(N^{-1/2})$. Also $v_1 + s = u_1(L') + u_1(L) + o(1) = 2u_1(\frac{\pi}{2}) + O(\delta) + o(1)$, and $\sin 2u_1(\frac{\pi}{2}) = \frac{2\sqrt{a/b}}{1 + a/b} = r_{ab}$, and $r_{ab} \ge 0.79$ for central splits once k is large (the limit value at $p = \epsilon k$ is 0.8). Hence $\mathbb{E}\sin(v + s) = r_{ab}(1 + o(1))$. (d) Upper bound: $\log M(\frac{\pi}{2} + \delta) \le \log M(\frac{\pi}{2}) + \delta D_1$

by concavity. Also $M(\frac{\pi}{2}-\delta) = \cos^b\delta\, M(\frac{\pi}{2})\,\mathbb{E}_0(1-tT)_+^b$, and on $\{T \le 1/(2t)\}$, $(1-tT)^b \ge e^{-btT-bt^2T^2}$. By Jensen, with $b\mathbb{E}_0T = D_1$, $\mathbb{E}_0T^2 = \frac{p}{q-1} \le 2$, and $\mathbb{P}(T > \frac{1}{2t}) \le \mathbb{E}_0T^b(2t)^b$ super-polynomially small, we get $\log M(\frac{\pi}{2}-\delta) \ge \log M(\frac{\pi}{2}) - tD_1 - O(N\delta^2)$. Lower bound: by concavity $\log M(\frac{\pi}{2}-\delta) \le \log M(\frac{\pi}{2}) - \delta D_1$; and as in Proposition 19, $\log M(\frac{\pi}{2}+\delta) \ge \log M(\frac{\pi}{2}) + tD_1 - O(N\delta^2)$. Since $t-\delta = O(\delta^3)$, (d) follows; the bound on D_1 is Lemma 17. □

Proof of Theorem 7. By Theorem 3, a split is safe at L iff $\log I_{\text{out}} > \log I_{\text{in}}$. For central splits, by Lemma 27, this is iff $\Psi < \log(2r_{ab}\sqrt{\pi N}) + o(1)$. Put $\delta = (\log k + c)/(4k)$ and $r = 2\sqrt{pq}/k \in [0.8, 1]$. Then $r_{ab} = r(1+O(1/k))$, $N = 2k-2$, and $N\delta^2 = o(1)$. So the condition reads

$$(*) \qquad \tfrac{r}{2}(\log k + c) < \tfrac{1}{2}\log k + \tfrac{1}{2}\log(8\pi) + \log r + o(1).$$

Upper bound. The balanced split ($p = \lfloor k/2 \rfloor$) has $r = 1 - O(k^{-2})$. By $(*)$ it is unsafe for $c > \log 8\pi$ and k large. Hence $\phi_k \le L(\lfloor \frac{k}{2} \rfloor, \lceil \frac{k}{2} \rceil) \le \frac{\pi}{2} + \frac{\log k + \log 8\pi + o(1)}{4k}$. *Lower bound, central splits.* Let $c = \log 8\pi - \epsilon'$. Using $\log r \ge -(1-r)/r \ge -\frac{5}{4}(1-r)$, the right side of $(*)$ minus the left side is at least $(1-r)\big(\frac{1}{2}\log k + \frac{1}{2}\log 8\pi - \frac{5}{4}\big) + \frac{r}{2}\epsilon' - o(1) \ge 0.4\epsilon' - o(1) > 0$. So every central split is safe for large k. *Lower bound, extreme splits.* The proof of Theorem 5(i) applies verbatim with $c = \log 8\pi$ in the range $y = \sqrt{pq}/k \le \sqrt{p/k} < \sqrt{\epsilon}$. There $h(\sqrt{\epsilon}) = (\sqrt{\epsilon} - \frac{1}{2})\log k + O(1) \to -\infty$ and $h(y_{\min}) \to -\frac{3}{2}\log 2$, so by convexity $\Phi < 0$ for large k, uniformly over extreme splits. Hence $\phi_k = \min_p L(p, k-p) \ge \frac{\pi}{2} + \frac{\log k + \log 8\pi - \epsilon'}{4k}$ for large k, for every $\epsilon' > 0$. □

7. All sufficiently large k: proof of Theorem 9

Write $L_{\text{bal}} = L(\lfloor \frac{k}{2} \rfloor, \lceil \frac{k}{2} \rceil)$. By symmetry we only consider $p \le q$.

Lemma 28 (far-from-balanced splits)**.** *There is K_A such that for $k \ge K_A$ every split with $p \le q$ and $p \le 0.41k$ satisfies $L(p, k-p) > L_{\text{bal}}$.*

Proof. By the upper-bound part of the proof of Theorem 7, $L_{\text{bal}} < \frac{\pi}{2} + \frac{\log k + \log 8\pi + 1/40}{4k}$ for large k. Let $c_+ = \log 8\pi + \frac{1}{20}$ and $\delta_+ = (\log k + c_+)/(4k)$. For a central split ($p \ge k/5$) with $p \le 0.41k$ we have $r = 2\sqrt{pq}/k \in [0.8, 0.984]$. At $c = c_+$ the right side of $(*)$ minus the left side equals $\frac{1-r}{2}(\log k + \log 8\pi) - \frac{r}{40} + \log r \ge 0.008\log k - \frac{1}{4}$. This tends to ∞, so the split is safe at δ_+ for large k. For extreme splits ($p < k/5$) we repeat the proof of Theorem 5(i) with $c = c_+$, in the range $y = \sqrt{pq}/k < 0.4$. The restriction $c < 2\log 2$ in that proof is used only at $y = \frac{1}{2}$, which lies outside this range. Here $h(0.4) = (0.4 - \frac{1}{2})\log k + O(1) \to -\infty$ and $h(y_{\min}) \to -\frac{3}{2}\log 2$. Since h is convex, $\Phi \le \max h + \frac{1}{2}\log\frac{3}{2} + o(1) < 0$, and Proposition 25 shows these splits are safe at δ_+. In both cases $L(p,q) > \frac{\pi}{2} + \delta_+ > L_{\text{bal}}$. □

Setting for near-balanced splits. For the rest of this section $a, b > 0$ are real, $N = a+b$, $x = a/N \in [0.4, 0.6]$, and $\ell \in \Lambda := [\frac{\pi}{2} - \frac{1}{10}, \frac{\pi}{2} + \frac{1}{10}]$. (For the splits of k we have $N = 2k-2$.) Let $\mu = \mu_{a,\ell}$ be the probability measure on $(0,\ell)$ with density $A_\ell/M_{a,b}(\ell)$, where $A_\ell(u) = \sin^a u \sin^b(\ell-u) = e^{-V(u)}$. Write E, Var, Cov for its moments, u_1 for its mode ($V'(u_1) = 0$), and $\psi_\ell(u) = \log\sin u - \log\sin(\ell-u)$, so that $\partial_a V = -\psi_\ell$ when $b = N - a$ with N fixed, and $\partial_b V = -\log\sin(\ell-u)$ when a is fixed. Put $\Theta(u) = 1 + |\log\sin u| + |\log\sin(\ell-u)| + |\cot u| + |\cot(\ell-u)|$, $B = [u_1 - \frac{1}{4}, u_1 + \frac{1}{4}]$ and $B_1 = [u_1 - \frac{1}{14}, u_1 + \frac{1}{14}]$. Constants C, C_m, c_0 and the implied constants in $O(\cdot)$ do not depend on a, ℓ, N. All statements hold for $N \ge N_1$, where N_1 may depend on the number of moments used.

Lemma 29 (geometry)**.** *(i) u_1, $\ell - u_1 \in [0.58, 0.99]$. (ii) $V'' \ge N$ on $(0,\ell)$. (iii) On B: $\sin u, \sin(\ell-u) \ge 0.32$, $V'' \le 10N$, $|V^{(j)}| \le CN$ and $|\psi_\ell^{(j)}| \le C$ for $j \le 5$. (iv) On $(0,\ell)$: $|\psi_\ell^{(j)}| \le C\Theta^{j+1}$ and $|V^{(j)}| \le CN\Theta^j$ for $1 \le j \le 5$.*

Proof. u_1 solves $a\cot u = b\cot(\ell-u)$, and $0<\partial_\ell u_1<1$ (proof of Lemma 27(a)). At $\ell=\frac{\pi}{2}$, $\tan^2 u_1 = a/b \in [\frac{2}{3},\frac{3}{2}]$, so u_1 and $\frac{\pi}{2}-u_1$ lie in $[0.684, 0.887]$; this gives (i). (ii) holds because $\csc^2\ge 1$. On B both u and $\ell-u$ lie in $[0.33, 1.24]$, which gives (iii). (iv) holds because the derivatives of $\log\sin$ are polynomials in cot. □

Lemma 30 (tails and moments)**.** *(i) $Z := \int_0^\ell e^{-(V-V(u_1))}du \ge c_0 N^{-1/2}$, so the density of μ is at most $C\sqrt{N}$. (ii) $E|u-u_1|^m \le C_m N^{-m/2}$. (iii) $E[\Theta^m;\, u\notin B_1]\le C_m e^{-N/700}$.*

Proof. (i) On B, $V-V(u_1)\le 5N(u-u_1)^2$. (ii) By Lemma 29(ii), $V(u)-V(u_1)\ge \frac{N}{2}(u-u_1)^2$, so $E|u-u_1|^m \le Z^{-1}\int |y|^m e^{-Ny^2/2}dy$. (iii) Off B_1 we have $V-V(u_1)\ge N/392$. With $\theta=1/600$, $e^{-(V-V(u_1))}\le e^{-(1-\theta)N/392}e^{\theta V(u_1)}A_\ell^\theta$, and $e^{\theta V(u_1)}\le(\sin 0.58)^{-\theta N}\le e^{0.61\theta N}$. Hence the left side is at most $Z^{-1}e^{-0.00153N}\int_0^\ell \Theta^m A_\ell^\theta\,du$, and the last integral is bounded once $\theta a,\theta b\ge m+2$. □

Lemma 31 (Stein identities)**.** *(i) If $g\in C^1(0,\ell)$ and $|g|+|g'|\le C\Theta^m$, then $E\,g'(u)=E[g(u)V'(u)]$. (ii) Let $|c-u_1|\le C/N$ and $y=u-c$. Then $Ey=O(N^{-1})$, $Ey^2=V''(c)^{-1}+O(N^{-2})$, $Ey^3=O(N^{-2})$, and $E|y|^m=O(N^{-m/2})$.*

Proof. (i) Integrate by parts on $(\epsilon,\ell-\epsilon)$; the boundary terms vanish because e^{-V} vanishes to order $\ge 0.4N$ at 0 and ℓ. (ii) The moment bound follows from Lemma 30(ii). By Lemma 29, $|V'(c)|=|V'(c)-V'(u_1)|\le C$, and on B_1 $V'(u)=V'(c)+V''(c)y+\frac12 V'''(c)y^2+R$ with $|R|\le CN|y|^3$. The contributions of $u\notin B_1$ to the identities below are $O(e^{-N/800})$ by Lemma 30(iii), since $|V'|\le N\Theta$. Applying (i) with $g=1$, $g=y^2$ and $g=y$: $0=V'(c)+V''(c)Ey+O(NEy^2)$, so $Ey=O(N^{-1})$. Next $2Ey=V'(c)Ey^2+V''(c)Ey^3+O(NEy^4)$, so $Ey^3=O(N^{-2})$. Finally $1=V'(c)Ey+V''(c)Ey^2+\frac12V'''(c)Ey^3+O(NEy^4)$, so $Ey^2=V''(c)^{-1}+O(N^{-2})$. □

Lemma 32 (smooth functionals)**.** *Let $F,G\in C^3(0,\ell)$, possibly depending on a and ℓ, with $|F^{(j)}|,|G^{(j)}|\le C$ on B for $j\le 3$ and $|F|,|G|\le C\Theta^m$ on $(0,\ell)$. Then $EF=F(u_1)+O(N^{-1})$, $\operatorname{Cov}(F,G)=F'(u_1)G'(u_1)/V''(u_1)+O(N^{-2})$ (in particular $\operatorname{Cov}(F,G)=O(N^{-1})$ and $\operatorname{Var}F=F'(u_1)^2/V''(u_1)+O(N^{-2})$), and $\kappa(F,G,G):=E[(F-EF)(G-EG)^2]=O(N^{-2})$.*

Proof. Let $y=u-u_1$ and $Y=y-Ey$. By Lemma 31 with $c=u_1$: $Ey=O(N^{-1})$, $EY^3=Ey^3-3EyEy^2+2(Ey)^3=O(N^{-2})$, and $\operatorname{Var}y=V''(u_1)^{-1}+O(N^{-2})$. On B_1 write $F=F(u_1)+F'(u_1)y+R_F$ with $|R_F|\le Cy^2$, and $R_F=\frac12F''(u_1)y^2+R_F'$ with $|R_F'|\le C|y|^3$; similarly for G. Contributions from $u\notin B_1$ are exponentially small by Lemma 30(iii). The first claim follows. Next, $\operatorname{Cov}(F,G)=F'G'\operatorname{Var}y+F'\operatorname{Cov}(y,R_G)+G'\operatorname{Cov}(y,R_F)+\operatorname{Cov}(R_F,R_G)$. Here $\operatorname{Cov}(y,R_F)=\frac12F''\operatorname{Cov}(y,y^2)+\operatorname{Cov}(y,R_F')=O(N^{-2})$, since $\operatorname{Cov}(y,y^2)=Ey^3-EyEy^2$. The same holds for G, and $|\operatorname{Cov}(R_F,R_G)|\le(ER_F^2\,ER_G^2)^{1/2}=O(N^{-2})$. Finally $F-EF=F'(u_1)Y+Q_F$ with $|Q_F|\le C(y^2+N^{-1})$ on B_1, and similarly for G. Expanding, $\kappa=F'G'^2EY^3+E\,O(Y^2|Q|+|Y||Q|^2+|Q|^3)$, where $|Q|=|Q_F|+|Q_G|$, and this is $O(N^{-2})$. □

Lemma 33 (odd functions)**.** *Let $|c-u_1|\le C/N$. Suppose that for $|y|\le\frac{1}{15}$ we have $g(c+y)=-g(c-y)$ and $|g(c+y)|\le|y|^m$, and that $|g|\le C\Theta^{m'}$ on $(0,\ell)$. Then $Eg(u)=O(N^{-(m+1)/2})$.*

Proof. Let $\Xi(y)=V(c+y)-V(c)$, so $Eg=\int g(c+y)e^{-\Xi(y)}dy/\int e^{-\Xi(y)}dy$. For $|y|\le\frac{1}{15}$ we have $c+y\in B_1$. There $|\Xi'(0)|=|V'(c)|\le C$, $N\le\Xi''\le 10N$ and $|\Xi'''|\le CN$, so $-C|y|+\frac{N}{2}y^2\le\Xi(\pm y)\le C|y|+5Ny^2$ and $|\Xi(y)-\Xi(-y)|\le C(|y|+N|y|^3)$. Hence $\int e^{-\Xi}\ge c_0N^{-1/2}$. By oddness,

$$\Big|\int_{-1/15}^{1/15} g(c+y)e^{-\Xi(y)}dy\Big| = \Big|\int_0^{1/15} g(c+y)\big(e^{-\Xi(y)}-e^{-\Xi(-y)}\big)dy\Big|$$
$$\le C\int_0^\infty y^m(y+Ny^3)e^{Cy-Ny^2/2}dy = O(N^{-\frac{m+2}{2}}).$$

The rest of the domain contributes $O(e^{-N/800})\int e^{-\Xi}$: on B_1 because $\Xi(y) \geq -C|y| + \frac{N}{2}y^2$ with $|y| > \frac{1}{15}$, and off B_1 by Lemma 30(iii). □

Lemma 34 (the angular mean)**.** *Let $\ell = L$ and let $s = s(a) \in (0, L)$ be defined by $E\sin(u-s) = 0$, as in Theorem 3. Then $|s - u_1| = O(N^{-1})$, $s'(a) = \psi_L'(s)/V''(s) + O(N^{-2})$ (in particular $s' = O(N^{-1})$), and $s''(a) = O(N^{-2})$. Here derivatives are taken in a with $b = N - a$ and L fixed. Also $\partial_b s = O(N^{-1})$ when a and L are fixed.*

Proof. With $y_1 = u - u_1$, $s - u_1$ is an argument of Ee^{iy_1}, since $s = \arg Ee^{iu}$. Moreover $E\cos y_1 \geq \frac{1}{2}$, so $\operatorname{Re} Ee^{iy_1} > 0$, and $|s - u_1| < L < \frac{3\pi}{2}$. Hence $s - u_1$ is the principal argument, $\tan(s - u_1) = E\sin y_1 / E\cos y_1$, and $|s - u_1| \leq C|E\sin y_1| = O(N^{-1})$ because $E\sin y_1 = Ey_1 + O(E|y_1|^3)$. The a-derivative of the log-density of μ is $\psi - E\psi$, with $\psi = \psi_L$. Differentiating $E\sin(u - s(a)) = 0$ gives $s' = P/Q$ with $P = \operatorname{Cov}(\sin(u-s), \psi)$ and $Q = E\cos(u-s) = 1 + O(N^{-1})$. Put $y = u - s$ and Taylor-expand ψ at s on B_1:

$$P = (\psi(s) - E\psi)E\sin y + \psi'(s)E[y\sin y] + \tfrac{1}{2}\psi''(s)E[y^2\sin y] + O(Ey^4) = \psi'(s)Ey^2 + O(N^{-2}).$$

This uses $E\sin y = 0$, $E[y\sin y] = Ey^2 + O(Ey^4)$ and $E[y^2\sin y] = Ey^3 + O(E|y|^5) = O(N^{-2})$ (Lemma 31(ii) with $c = s$). Since $Ey^2 = V''(s)^{-1} + O(N^{-2})$, the formula for s' follows. Next, $P' = \kappa(\sin(u-s), \psi, \psi) - s'\operatorname{Cov}(\cos(u-s), \psi) = O(N^{-2})$ and $Q' = \operatorname{Cov}(\cos(u-s), \psi) + s'E\sin(u-s) = O(N^{-1})$, both by Lemma 32. So $s'' = (P'Q - PQ')/Q^2 = O(N^{-2})$. Finally, differentiating $E\sin(u-s) = 0$ in b gives $\partial_b s = \operatorname{Cov}(\sin(u-s), \log\sin(L-u))/Q = O(N^{-1})$. □

The three terms. Fix $L = \frac{\pi}{2} + \delta \in \Lambda$ with $\delta \geq 0$, and $L' = \pi - L$. Let $\mu = \mu_{a,L}$ (variable u) and $\mu' = \mu_{a,L'}$ (variable v), with expectations E and E'. Theorem 3, together with the substitution $v = \pi - u$ in I_{out}, gives $I_{\text{in}} = \frac{1}{2}M_{a,b}(L)\,E|\sin(u-s)|$ and $I_{\text{out}} = M_{a,b}(L')\,E'\sin(v+s)$. Hence

$$G(a) := \log I_{\text{in}} - \log I_{\text{out}} = T_1 + T_2 - T_3 - \log 2,$$

$$T_1 = \log\frac{M_{a,b}(L)}{M_{a,b}(L')}, \qquad T_2 = \log E|\sin(u-s)|, \qquad T_3 = \log E'\sin(v+s),$$

with $b = N - a$ and $s = s(a)$. Primes denote derivatives in a.

Proposition 35 (mass term)**.** $T_1'' \leq -4\delta/N + C(\delta^3 N^{-1} + N^{-2})$.

Proof. $\partial_a^2 \log M_{a,N-a}(\ell) = \operatorname{Var}_{\mu_{a,\ell}}(\psi_\ell)$. By Lemma 32 this equals $q(x,\ell)/N + O(N^{-2})$, where $q = \psi_\ell'(u_1)^2/K$ and $K = V''(u_1)/N = x\csc^2 u_1 + (1-x)\csc^2(\ell - u_1)$. Put $\gamma = \cot u_1$. The mode equation $x\cot u_1 = (1-x)\cot(\ell - u_1)$ gives $\psi_\ell'(u_1) = \gamma/(1-x)$, $K = 1 + x\gamma^2/(1-x)$ and $q = \gamma^2/\big((1-x)(1-x+x\gamma^2)\big)$. Hence $\partial q/\partial\gamma = 2\gamma/(1-x+x\gamma^2)^2$, $\partial_\ell u_1 = (1-x)\csc^2(\ell-u_1)/K$ and $\partial_\ell\gamma = -\csc^2 u_1\,\partial_\ell u_1$. At $\ell = \frac{\pi}{2}$ we have $\gamma^2 = (1-x)/x$, $K = 2$ and $\partial_\ell u_1 = \frac{1}{2}$, so

$$\partial_\ell q(x, \tfrac{\pi}{2}) = -\frac{\gamma}{2(1-x)^2}\cdot\frac{1}{2x} = -\frac{1}{4\,(x(1-x))^{3/2}} \leq -2.$$

Since q is real-analytic on $[0.4, 0.6] \times \Lambda$, $q(x,L) - q(x,L') = 2\delta\,\partial_\ell q(x, \frac{\pi}{2}) + O(\delta^3)$, and $T_1'' = (q(x,L) - q(x,L'))/N + O(N^{-2})$. □

Proposition 36 (outer term)**.** $|T_3''| \leq CN^{-2}$.

Proof. Let $F_a(v) = \sin(v + s(a))$ and $\chi = \psi_{L'}$. On the set B of μ' we have $v + s \in [0.9, 2.24]$ by Lemmas 29(i) and 34, so $F_a \geq 0.78$ there and $E'F_a \geq \frac{1}{2}$. The a-derivative of the log-density of μ' is $\chi - E'\chi$. Therefore

$$(E'F_a)' = \operatorname{Cov}'(F_a, \chi) + s'E'\cos(v+s),$$
$$(E'F_a)'' = \kappa'(F_a, \chi, \chi) + 2s'\operatorname{Cov}'(\cos(v+s), \chi) + s''E'\cos(v+s) - s'^2E'F_a.$$

By Lemmas 32 and 34, $(E'F_a)' = O(N^{-1})$ and $(E'F_a)'' = O(N^{-2})$, so $T_3'' = (E'F_a)''/E'F_a - ((E'F_a)'/E'F_a)^2 = O(N^{-2})$. □

Proposition 37 (inner term)**.** $|T_2''| \le CN^{-2}$.

Proof. We work in the frame that moves with s. Let $q_a(y) = A_L(s(a)+y)/M_{a,b}(L)$ for $y \in (-s, L-s)$, extended by 0; this is the density of $y = u - s$. Then $h(a) := E|\sin(u-s)| = \int |\sin y|\, q_a(y)\, dy$, and since $\partial_a V = -\psi$ and $\partial_a \log M_{a,b}(L) = E\psi$,

$$S_a(y) := \partial_a \log q_a(y) = \psi(s+y) - E\psi - s'V'(s+y),$$
$$\partial_a S_a(y) = 2s'\psi'(s+y) - \operatorname{Var}\psi - s''V'(s+y) - s'^2V''(s+y).$$

A_L vanishes to order $\ge 0.4N$ at 0 and L. Hence $q_a(y)$ and its first two a-derivatives (which equal q_aS_a and $q_a(S_a^2 + \partial_a S_a)$ inside the support) extend continuously by 0. They are bounded uniformly on $\{|y| \le L\} \times [a_0 - \eta, a_0 + \eta]$ for small $\eta > 0$, so we may differentiate under the integral sign. This gives $E_q[S_a] = 0$ and $E_q[S_a^2 + \partial_a S_a] = 0$ (differentiate $\int q_a = 1$ twice), together with

$$(\log h)'' = \operatorname{Var}_w(S_a) + E_w[\partial_a S_a], \qquad E_w[f] := E_q[|\sin y|\, f]/h.$$

Size of S_a. By Lemma 32, $E\psi = \psi(u_1) + O(N^{-1})$. By Lemma 34, $|s - u_1| = O(N^{-1})$, $|V'(s)| \le C$ and $s' = \psi'(s)/V''(s) + O(N^{-2})$. Hence $S_a(0) = O(N^{-1})$, $\partial_y S_a(0) = \psi'(s) - s'V''(s) = O(N^{-1})$, and $\partial_y^2 S_a = \psi'' - s'V''' = O(1)$ on $B_1 - s$. So $|S_a(y)| \le C(N^{-1} + y^2)$ for $s + y \in B_1$, while $|S_a| \le CN\Theta(s+y)^2$ everywhere. *Lower bound for h.* The density of u is at most $C\sqrt{N}$ (Lemma 30(i)). With $t = c_0 N^{-1/2}$ small enough we get $\mathbb{P}(|y| \le t) \le \frac{1}{2}$. Since $|y| < L < 1.7$ we have $|\sin y| \ge \frac{1}{2}\min(|y|, 1)$, hence $h \ge \frac{1}{2}E\min(|y|,1) \ge \frac{t}{4} \ge cN^{-1/2}$. *Variance part.* $\operatorname{Var}_w(S_a) \le E_q[|\sin y| S_a^2]/h \le C\, E_q[|y|(N^{-1} + y^2)^2]/h + O(e^{-N/800}) = O(N^{-2})$. *Drift part.* Let $D(y) = \partial_a S_a(y) - \partial_a S_a(0)$. Since $E_w - E_q$ annihilates constants, $E_w[\partial_a S_a] = E_q[\partial_a S_a] + (E_w - E_q)[D]$, and $E_q[\partial_a S_a] = -E_q[S_a^2] = O(N^{-2})$. On $B_1 - s$, using $s' = O(N^{-1})$, $s'' = O(N^{-2})$ and Lemma 29(iii), we have $D(y) = \beta y + D_2(y)$ with $\beta = 2s'\psi''(s) - s''V''(s) - s'^2V'''(s) = O(N^{-1})$ and $|D_2(y)| \le Cy^2/N$. Everywhere $|D| \le C\Theta(s+y)^4$. Moreover $E_q y = Eu - s = O(N^{-1})$, and $E_w y = E[|\sin(u-s)|(u-s)]/h = O(N^{-3/2})/h = O(N^{-1})$ by Lemma 33 with $m = 2$. Also $E_w y^2 + E_q y^2 = O(N^{-1})$. Hence $(E_w - E_q)[D] = \beta(E_w y - E_q y) + O(N^{-2}) = O(N^{-2})$. So $(\log h)'' = O(N^{-2})$, and $T_2 = \log h$. □

Proof of Theorem 9. Let $L = L_{\mathrm{bal}}$ and $\delta = L - \frac{\pi}{2}$. Then $\delta \ge \phi_k - \frac{\pi}{2} > \frac{\log k}{4k}$ for large k by Theorem 5(i). The upper-bound part of the proof of Theorem 7 gives $\delta \le \frac{\log k + \log 8\pi + 1}{4k} \le \frac{2\log k}{k}$ for large k, so $L \in \Lambda$. By Lemma 28 it remains to treat splits with $0.41k < p \le q$. For these $a = 2p - 1 \in [0.4N, 0.6N]$ when $k \ge 10$. By Propositions 35–37, for all real $a \in [0.4N, 0.6N]$,

$$G''(a) \le -\frac{4\delta}{N} + C\Big(\frac{\delta^3}{N} + \frac{1}{N^2}\Big) \le -\frac{\log k}{kN} + \frac{C'}{N^2} < 0 \qquad (k \ge K_B).$$

Moreover $G(a) = G(N-a)$. Indeed, $u \mapsto L - u$ and $s \mapsto L - s$ exchange a and b in I_{in}, and $u \mapsto \pi + L - u$ does so in I_{out}. The balanced exponents $a_{\mathrm{bal}} = 2\lfloor \frac{k}{2} \rfloor - 1$ and $N - a_{\mathrm{bal}}$ satisfy $G = 0$ at L. If k is even, then $a_{\mathrm{bal}} = \frac{N}{2}$ is the maximum point of the strictly concave symmetric function G, so $G(a) < 0$ for every other a in the window. If k is odd, then $G(\frac{N}{2} \pm 1) = 0$ and strict concavity gives $G < 0$ outside $[\frac{N}{2} - 1, \frac{N}{2} + 1]$; every other admissible $a = 2p - 1$ satisfies $|a - \frac{N}{2}| \ge 3$. In both cases every non-balanced split in the window has $I_{\mathrm{out}} > I_{\mathrm{in}}$ at L, i.e. $\Delta_{p,q}(L) > 0$, and hence $L(p,q) > L_{\mathrm{bal}}$ by Theorem 3. Take $K_0 = \max(K_A, K_B, 10)$. □

Monotonicity.

Proposition 38 (derivatives in b)**.** *Let $L = \frac{\pi}{2} + \delta \in \Lambda$ with $\delta \geq 0$, and let T_1, T_2, T_3 be as above, now regarded as functions of b with a and L fixed. Then $\partial_b T_1 \geq 0.22\,\delta - C\delta/N$ and $|\partial_b T_2| + |\partial_b T_3| \leq C/N$.*

Proof. T_1. We have $\partial_b \log M_{a,b}(\ell) = m_\ell := E_\ell \log\sin(\ell - u)$, and $\partial_b T_1 = m_L - m_{L'}$. The log-density of $\mu_{a,b,\ell}$ has ℓ-derivative $b\cot(\ell-u) - E\,b\cot(\ell-u)$, and A_ℓ vanishes to order b at $u = \ell$. Hence $\partial_\ell m_\ell = E\cot(\ell-u) + b\,\mathrm{Cov}(\log\sin(\ell-u), \cot(\ell-u))$. By Lemma 32, and since $\partial_\ell u_1 = b\csc^2(\ell-u_1)/V''(u_1)$ (proof of Proposition 35),

$$\partial_\ell m_\ell = \cot(\ell-u_1) - \frac{b\cot(\ell-u_1)\csc^2(\ell-u_1)}{V''(u_1)} + O(N^{-1}) = \cot(\ell-u_1)\,(1-\partial_\ell u_1) + O(N^{-1}).$$

By Lemma 29(i), $\cot(\ell-u_1) \geq \cot 0.99 \geq 0.65$. Also $1 - \partial_\ell u_1 = x\csc^2 u_1/K$, and this is at least $0.4\csc^2(0.99)/\csc^2(0.58) \geq 0.17$. So $\partial_\ell m_\ell \geq 0.11 - C/N$ on Λ, and $m_L - m_{L'} \geq 2\delta(0.11 - C/N)$.
T_3. The b-derivative of the log-density of μ' is $\log\sin(L'-v) - E'\log\sin(L'-v)$. So $\partial_b E'\sin(v+s) = \mathrm{Cov}'(\sin(v+s), \log\sin(L'-v)) + \partial_b s\,E'\cos(v+s) = O(N^{-1})$ by Lemmas 32 and 34, and $E'\sin(v+s) \geq \frac{1}{2}$.
T_2. As in Proposition 37, $\partial_b \log h = E_w[S^b]$ with $S^b(y) = \log\sin(L-s-y) - E\log\sin(L-u) - \partial_b s\,V'(s+y)$. By Lemmas 32 and 34, $S^b(0) = O(N^{-1})$, $\partial_y S^b(0) = O(1)$ and $|\partial_y^2 S^b| \leq C$ on $B_1 - s$. Hence $E_w[S^b] = S^b(0) + \partial_y S^b(0)\,E_w y + O(E_w y^2) + O(e^{-N/800}) = O(N^{-1})$, because $E_w y = O(N^{-1})$ (Lemma 33) and $E_w y^2 = O(N^{-1})$. □

Proof of Theorem 10. Let (p,q) be a split of k with $0.42k \leq p \leq 0.58k$, and put $a = 2p-1$, $b_0 = 2q-1$, $L = L(p,q)$, $\delta = L - \frac{\pi}{2}$. Then $\delta \geq \phi_k - \frac{\pi}{2} > \frac{\log k}{4k}$ for large k (Theorem 5(i)). Also $\delta \leq \frac{\log k}{k}$ for large k. Indeed, apply Lemma 27 to $(\min(p,q), \max(p,q))$, using $L(p,q) = L(q,p)$. At $\delta = \frac{\log k}{k}$ it gives $\Psi = 2r\log k + o(\log k)$, which exceeds $\log(2r_{ab}\sqrt{\pi N}) + o(1)$, so the split is unsafe there. For $b \in [b_0, b_0+2]$ we have $a/(a+b) \in [0.4, 0.6]$ once k is large. Let $g(b) = \log I_{\mathrm{in}} - \log I_{\mathrm{out}}$, computed at L with exponents (a,b). Then $g(b_0) = 0$, and by Proposition 38,

$$g'(b) = \partial_b T_1 + \partial_b T_2 - \partial_b T_3 \geq 0.22\,\delta - \frac{C}{N} > \frac{0.05\log k}{k} - \frac{C}{2k-2} > 0 \qquad (k \geq K_2').$$

So $g(b_0+2) > 0$, i.e. $\Delta_{p,q+1}(L) < 0$, and $L(p,q+1) < L$ by Theorem 3. This proves endpoint monotonicity for $k \geq K_2'$. Now take $K_2 = \max(K_0, K_2')$. For $k \geq K_2$ we have $\phi_k = L(p_*, q_*)$ with $(p_*, q_*) = (\lfloor \frac{k}{2} \rfloor, \lceil \frac{k}{2} \rceil)$ (Theorem 9). Since (p_*, q_*+1) is a split of $k+1$, $\phi_{k+1} \leq L(p_*, q_*+1) < L(p_*, q_*) = \phi_k$. □

Numerical check of the three terms (`numerics/bln_decomp.py`, `numerics/decomp.txt`). At $a = \frac{N}{2}$ and $L = L_{\mathrm{bal}}$, as N runs from 40 to 1280: $N^2 T_1''$ runs from -11.7 to -19.3, against $-4N\delta$ from -12.1 to -19.4; $N^2 T_2''$ runs from 0.06 to 0.005; $N^2 T_3'' \to -4$; and $N^2 G''$ runs from -7.9 to -15.3. These are consistent with the propositions above. For Proposition 38 (`numerics/bln_decomp_b.py`, `numerics/decomp_b.txt`), at the balanced split for $k = 20, \ldots, 640$: $N\partial_b T_1$ agrees with $2N\delta \cdot \frac{1}{2}\sqrt{x/(1-x)}$ to three decimals (from 2.98 to 4.84), $N\partial_b T_2 \to -\frac{1}{2}$, and $N\partial_b T_3 \to 0$. These numbers concern only near-balanced splits. They say nothing about the size of K_0 or K_2, which also depend on the $o(1)$ terms in Lemma 28, and we did not make the constants explicit.

8. Rigorous verification for $k \leq 440$: proof of Theorem 8

Fix a split with $a = 2p-1$, $b = 2q-1$, and a rational $L \in (0, \pi)$. Expanding by the binomial theorem,

$$P(u) := \sin^a u\,\sin^b(u-L) = (2i)^{-a-b} \sum_{j=0}^{a}\sum_{l=0}^{b} \binom{a}{j}\binom{b}{l} (-1)^{a+b-j-l} e^{-i(2l-b)L}\, e^{i(2j-a+2l-b)u}.$$

Put $F_1(t) = \int_0^t P \sin$ and $F_2(t) = \int_0^t P \cos$. These are finite sums of $\int_0^t e^{inu}du = (e^{int}-1)/(in)$ with n odd. In the notation of the proof of Theorem 3, $F = \cos s\, F_1 - \sin s\, F_2$. Since $P < 0$ on $(0, L)$, $s = \operatorname{atan2}(-F_1(L), -F_2(L))$, and

$$\Delta_{p,q}(L) = F(\pi) - F(s) = \cos s\,(F_1(\pi) - F_1(s)) - \sin s\,(F_2(\pi) - F_2(s)).$$

Every quantity here is evaluated in Arb ball arithmetic (python-flint, precision $200 + 6k$ bits, which absorbs the 2^{a+b} cancellation), giving a rigorous enclosure of $\Delta_{p,q}(L)$. By the sign pattern of Theorem 3, a certified $\Delta_{p,q}(L) > 0$ proves $L(p,q) > L$, and a certified $\Delta_{p,q}(L) < 0$ proves $L(p,q) < L$.

For each k we take a rational L^* (denominator $\le 10^{12}$) halfway between the floating-point values of L_{bal} and $\min_{p<\lfloor k/2\rfloor} L(p, k-p)$. We certify $\Delta_{\mathrm{bal}}(L^*) < 0$ and $\Delta_{p,k-p}(L^*) > 0$ for every other $p \le k/2$, which gives $L_{\mathrm{bal}} < L^* < L(p, k-p)$. The other comparisons in Theorem 8 use separating rational points in the same way. The enclosures of ϕ_k come from certified signs at rational points within $5 \cdot 10^{-15}$ of $L_f \pm 10^{-12}$. The floating-point values only choose the test points; correctness rests on the certified signs. Code: `numerics/bln_rigorous.py`; the run for $k \le 200$ (all three claims) takes about 50 minutes on one core (output `numerics/rigorous_200.txt`), and the run `bln_rigorous.py 201 300` (the conjecture for $201 \le k \le 300$ and $\phi_{k+1} < \phi_k$ for $200 \le k < 300$) about 40 minutes (`numerics/rigorous_201_300.txt`). The range $301 \le k \le 400$ was run in four chunks, `bln_rigorous.py` with arguments 301 337, 338 363, 364 383 and 384 400, in parallel. Each chunk also certifies $\phi_{k+1} < \phi_k$ across its lower seam. This took about 85 minutes of wall time, at roughly 80 seconds per k near $k = 350$ (`numerics/rigorous_301_400.txt`). Near the balanced split the quantities compared are of size about $2^{-(a+b)/2}$, so the rational test points are converted to balls at 4000 bits; with fewer bits the first k of a run above $k \approx 370$ was not certified. The range $401 \le k \le 440$ (and $\phi_{k+1} < \phi_k$ for $400 \le k < 440$) was run as `bln_rigorous.py 401 440` on one core, at about 12 minutes per k (`numerics/rigorous_401_440.txt`).

Two further certificates. Theorems 5(ii) and 6 use Theorem 8 for $k \le 300$. Above that, on a finite range, they use the explicit closed forms of Propositions 25 and 19. These are certified in Arb by `numerics/bln_lower_arb.py` (all 99977500 pairs (k,p) with $301 \le k \le 2 \cdot 10^4$; output `numerics/lower_arb.txt`) and by `numerics/bln_odd_upper_arb.py` (odd $301 \le k \le 200001$; output `numerics/odd_upper_arb.txt`). Both scripts read the enclosures of Theorem 8 from the output files above. `bln_rigorous.py` certifies $\phi_k \in (L_f - 1.01 \cdot 10^{-12}, L_f + 1.01 \cdot 10^{-12})$ and prints L_f rounded to 12 decimals, so the scripts use the printed value $\pm 2 \cdot 10^{-12}$.

9. Numerical evidence

Computation of $L(p,q)$ (`numerics/bln_fast.py`). The quantities s, I_{in} and I_{out} of Theorem 3 are integrals of positive functions. We evaluate them in log scale with 1200-point Gauss–Legendre rules split at the peaks, and solve $\log I_{\mathrm{out}} = \log I_{\mathrm{in}}$ by Brent's method. Doubling the number of nodes changes L by at most $6 \cdot 10^{-14}$. The method reproduces $\phi_2 = 2\pi/3$, $\phi_3 = 1.962719002$ and $\phi_4 = 1.870658532$ to all digits.

Results (`numerics/sweep_300.txt`). For every $2 \le k \le 300$: the minimizing split is the balanced one (Conjecture 2, both parities); $p \mapsto L(p, k-p)$ is strictly decreasing on $1 \le p \le k/2$, with the smallest gap $1.3 \cdot 10^{-7}$; and ϕ_k is strictly decreasing in k. Conjecture 11 is rigorously verified for $p + q < 200$ (Theorem 8).

k	ϕ_k (numerical)	$4k(\phi_k - \frac{\pi}{2}) - \log k$	lower (Prop. 25)	upper (Prop. 15)
10	1.7115206092	3.3264	+0.457	3.383
20	1.6493102586	3.2854	+0.923	3.312
100	1.5904114962	3.2409	+1.242	3.246
300	1.5782417442	3.2307	+1.322	3.232
10^4	–	–	+1.382	3.2245

The last two columns are in the normalization $4k\delta - \log k$. The first gives the largest c such that $\Phi < 0$ for all splits at $\delta = (\log k + c)/(4k)$. The second gives the smallest δ satisfying the sufficient inequality $2k \log\tan\alpha > \log(2kB(\frac{1}{2}, k)/c)$ in the proof of Theorem 4(i), for even k. The limit $\log(8\pi) = 3.22417$ is matched by the heuristic Laplace expansion of the criterion. For a split with $\rho = p/k$ and $r = 2\sqrt{\rho(1-\rho)}$, one has $I_{\text{in}} \approx M(L)/(2\sqrt{\pi N})$ ($N = 2k-2$), $I_{\text{out}} \approx rM(\pi - L)$, and $\log M(\frac{\pi}{2} + \delta) - \log M(\frac{\pi}{2} - \delta) \approx 4\sqrt{pq}\,\delta$. This predicts $L(p,q) \approx \frac{\pi}{2} + \frac{\log k + \log 8\pi + 2\log r}{4kr}$, which is minimized at $r = 1$ on the admissible range $r \geq 2/\sqrt{k}$. The predicted gaps $L(p, k-p) - L(\frac{k}{2}, \frac{k}{2}) \approx (\frac{\log k + \log 8\pi}{2} - 1)\frac{(\rho - 1/2)^2}{k}$ match the observed ones (for example $1.1 \cdot 10^{-7}$ predicted against $1.3 \cdot 10^{-7}$ observed at $k = 300$). The proof of Theorem 9 does not use this expansion. Instead it controls the second derivative of the criterion in the exponent (§7).

References

[1] A. Barvinok, I. Novik, *A centrally symmetric version of the cyclic polytope*, Discrete Comput. Geom. 39 (2008) 76–99.

[2] A. Barvinok, S. J. Lee, I. Novik, *Neighborliness of the symmetric moment curve*, Mathematika 59 (2013) 223–249; arXiv:1104.5168.

[3] R. Sanyal, F. Sottile, B. Sturmfels, *Orbitopes*, Mathematika 57 (2011) 275–314.

[4] C. Vinzant, *Edges of the Barvinok–Novik orbitope*, Discrete Comput. Geom. 46 (2011) 479–487.

[5] J. G. Wendel, *Note on the gamma function*, Amer. Math. Monthly 55 (1948) 563–564.

[6] A. Prékopa, *On logarithmic concave measures and functions*, Acta Sci. Math. (Szeged) 34 (1973) 335–343.

[7] J. de Dios Pont, P. Ivanisvili, J. Madrid, *A new proof of the description of the convex hull of space curves with totally positive torsion*, Michigan Math. J. 76 (2026) 1043–1089.